

Comparing AI chatbots and web-based synchronous collaborative writing tools, leveraging group cohesion, motivation, and performance

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ABSTRACT

This study examines the effects of generative AI (GenAI) chatbots versus traditional web-based resources (TWRs) on group cohesion, motivation, and writing performance during face-to-face synchronous collaborative writing (SCW) tasks among L2 learners. Despite growing adoption of AI in language education, key gaps remain: limited psychological research on AI-supported collaborative writing, scarce attention to motivation and self-efficacy in group contexts, and absence of direct comparisons between AI chatbots and conventional tools in real-time, in-class settings. Sixty-three English-major sophomore students at a Chinese university were randomly assigned to two classes: an experimental cohort ($n = 33$) using Kimi AI chatbot and a control cohort ($n = 30$) using TWRs (e.g., dictionaries, search engines). Both groups worked in teams of 5–6 on a shared cloud document to revise a 500-word descriptive essay. Mixed-methods data included screen recordings, writing assessments, typing speed measures, pre/post motivation, perception questionnaires, and individual notes on team cohesion. Quantitative results showed the AI group completed writing and proofreading significantly faster (EWT: $M = 24:20$ vs. $41:10$; EPT: $M = 9:33$ vs. $15:25$), achieved higher essay scores ($M = 90.28$ vs. 81.11 , $p = .001$, $d = 2.43$), and reported substantially greater post-task motivation (ANCOVA, $p < .001$, $\eta^2 = 0.888$). Qualitative analysis of the cohesion notes suggested that AI helped streamline communication, reduce information overload, and support engagement, but it also introduced challenges, as its rapid and uniform responses could reduce discussion, limit exploration of alternatives, and encourage over-reliance on the tool. From a Group Cohesion Theory (GCT) perspective, AI appeared to strengthen task cohesion more than social cohesion, improving coordination and performance, though it did not always deepen peer interaction or shared meaning-making.

1. Introduction

Educational instruction has significantly benefited from technological advancement over the past decade. The ubiquity of technology in language classrooms has reinforced the usefulness of models like technology-enhanced language learning, mobile-assisted language learning, synchronous collaborative writing (SCW), and computer-assisted language learning, which have received increased scholarly and pedagogical attention (Marzuki et al., 2023). With the introduction of GenAI chatbots in late 2022, instructors have integrated large language models (LLMs) into course designs, with benefits and challenges for writing skill development (Marzuki et al., 2023).

Current research reveals a significant disconnect between AI implementation and the psychological factors (like motivation and self-efficacy) that drive successful collaborative writing (Liang et al., 2023; Xu et al., 2023). While pre-GenAI studies focused on pedagogical outcomes, they largely ignored affective variables (Li & Zhang, 2023).

Recent GenAI studies suggest that personalized feedback can increase individual motivation to write (Alghasab, 2025), yet this has not been sufficiently tested in SCW. Because motivation is the primary mediator of engagement and persistence during group negotiation (Lai & Yang, 2024), GenAI could, in theory, amplify collective efficacy by reducing anxiety and streamlining resource pooling. However, researchers have not compared AI chatbots against traditional web-based resources (TWRs) in face-to-face settings, while simultaneously measuring motivation, group cohesion, and final performance.

While prior research has documented performance advantages in AI-supported collaborative writing and isolated motivational benefits from chatbots (Li et al., 2025; Li & Tekwa, 2025), fewer studies articulate the underlying psychological mechanisms in group contexts. To address this gap, the present study draws on Group Cohesion Theory (GCT) (Carron et al., 1985; Festinger et al., 1950). GCT is particularly well-suited for this study. Developed in sport and organizational psychology (Carron et al., 1985), GCT provides a multidimensional framework that clearly

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distinguishes task cohesion, the group's shared commitment to goals and coordinated effort (from social cohesion), and the interpersonal bonds among members (Carron & Brawley, 2000). This distinction is highly relevant to synchronous collaborative writing, where teams must manage both linguistic demands and relational dynamics under time constraints (Heuzé et al., 2006). Unlike broader socio-cultural theories focused on long-term identity, GCT is sensitive to immediate, task-oriented processes and has strong predictive power for performance outcomes via mechanisms such as collective efficacy and reduced process loss. It therefore enables a nuanced analysis of how GenAI may enhance coordination while potentially limiting deeper peer negotiation in L2 classroom settings.

GenAI provides centralized, uniform, rapid, and interactive support (e.g., consistent suggestions for vocabulary, collocation, tone, and structure), which is hypothesized to enhance task cohesion more effectively than fragmented TWRs, leading to divergent information and prolonged debates. In contrast, AI could reduce coordination costs, minimize information overload, and promote swift consensus on content decisions, fostering stronger positive interdependence, with individuals perceiving that their contributions facilitate shared writing goals (Johnson & Johnson, 2009). This elevated task cohesion is expected to translate into motivational gains, including heightened collective efficacy (Bandura, 1997), a group's shared belief in its conjoint capability to succeed, which sustains engagement, reduces frustration from impasses, and boosts intrinsic motivation and persistence.

Consequently, stronger task cohesion and resulting motivational processes should predict superior performance outcomes, including efficiency (shorter writing/proofreading times) and improved writing quality. Social cohesion may also benefit indirectly from reduced conflict and more supportive peer exchanges enabled by AI's role as a neutral team member.

Fig. 1 illustrates this proposed explanatory model. This study examines whether and how these mechanisms unfold in face-to-face, in-class SCW by comparing Kimi AI to TWRs among L2 English majors, using mixed-methods data (screen recordings, assessments, questionnaires, and cohesion notes) to capture behavioral traces, perceptions,

and processes.

The explanatory model (Fig. 1) serves as the central theoretical foundation for this study. It was developed through the iterative integration of three established frameworks to articulate the precise mechanisms by which GenAI influences group dynamics in synchronous collaborative writing. GCT (Carron et al., 1985) supplied the core distinction between task cohesion and social cohesion and has been widely validated in performance-oriented group contexts. Social interdependence theory (Johnson & Johnson, 2009) posits positive interdependence as a mechanism that explains how tool features shape coordination and resource sharing. Collective efficacy (Bandura, 1997) then served as the motivational bridge linking cohesion to sustained effort and performance outcomes.

We constructed the model by mapping the distinctive characteristics of AI support (centralized, rapid, and uniform responses) onto these frameworks, producing the following causal pathway: AI support characteristics (centralized, rapid, uniform) → reduced coordination costs and information divergence → stronger task cohesion (faster consensus, clearer role integration) → elevated collective efficacy → higher motivation and persistence → superior performance (efficiency + quality). Social cohesion is expected to benefit indirectly but may be constrained if AI reduces opportunities for extended negotiation. This model extends prior GCT applications to technology-mediated learning by specifying tool features as antecedents of cohesion types—an extension largely absent from existing SCW literature.

2. Theoretical background and literature review

2.1. Group Cohesion Theory (GCT)

Festinger et al. (1950) originally defined group cohesion as forces binding members through shared goals, interpersonal attraction, and commitment. The concept evolved into a multidimensional model distinguishing task cohesion (goal-oriented collaboration) from social cohesion (personal bonds) (Aoyagi et al., 2008; Casey-Campbell & Martens, 2009). In sports, military, and organizational settings,

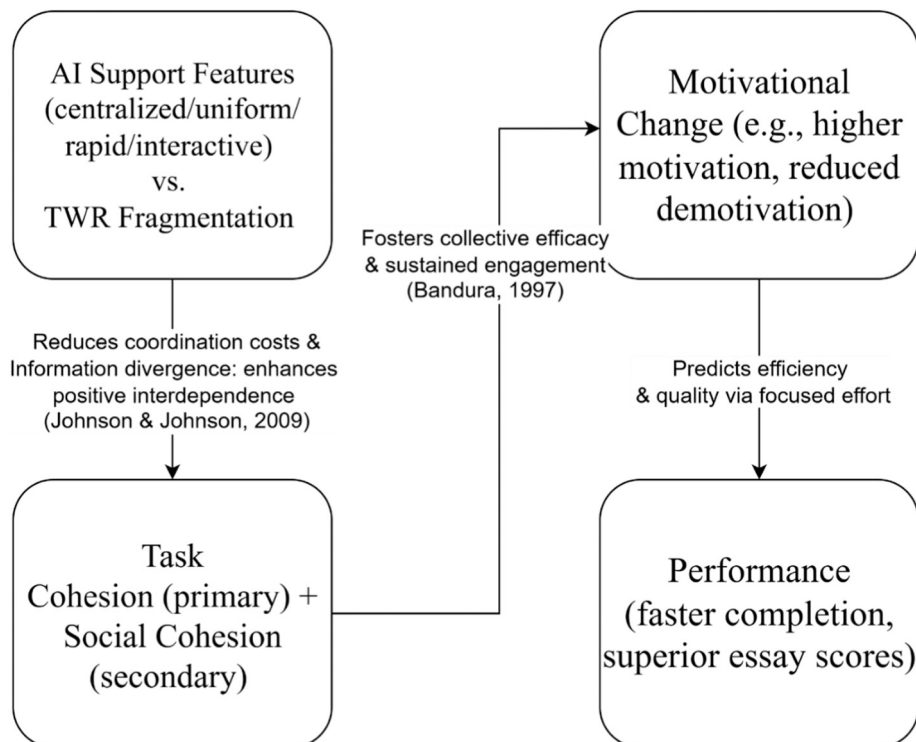


Fig. 1. Proposed explanatory model.

stronger cohesion predicts better performance, often mediated by collective efficacy and citizenship behaviors (García-Guiu et al., 2016).

In education, cohesion fosters experiential and relational growth, though results vary: some studies show post-intervention declines (Steen et al., 2014), while others highlight predictors like self-disclosure and gender (Zhu et al., 2024), or links to emotional intelligence, culture, and engagement (Fjermestad et al., 2024).

Technology-mediated collaborative learning introduces complexities. Early research noted that online settings promote reflection but may limit immediate bonding compared to face-to-face (Altebarmakian & Alterman, 2019). On asynchronous platforms, social presence and interaction quality enhance cohesion and efficacy (Dewiyanti et al., 2007). Synchronous and hybrid contexts remain underexplored, though real-time negotiation may particularly strengthen task cohesion (Tao et al., 2022). GenAI tools like Kimi AI can support collective efficacy and synchronous collaborative writing via interactive refinement, but risk reducing human interaction (Li et al., 2025; Li & Tekwa, 2025).

In second-language collaborative writing, process-oriented assessment, peer feedback, instructor support, and computer-mediated pre-writing improve collaboration, cohesion, and satisfaction (Chen & Cui, 2022; Ebadijalal & Moradkhani, 2025; Kaur & Chowdhury, 2022; Zhang & Chen, 2022). However, motivation, self-efficacy, and AI's influence on real-time cohesion in hybrid settings remain understudied (cf. Li & Zhang, 2023). Tools that reduce coordination barriers (e.g., uniform resources minimizing divergence) should amplify task cohesion by fostering positive interdependence (Johnson & Johnson, 2009; van den Hout et al., 2018), thereby enhancing motivation in synchronous collaborative writing (SCW).

Analytical lens and construct mapping: In this study, GCT serves as the descriptive backdrop and analytical perspective that guides the research design and interpretation. Building on models that distinguish task cohesion (group integration) from social cohesion (individual attraction), this study maps these constructs onto observable behaviors in face-to-face synchronous collaborative writing (SCW). Task cohesion is measured by goal-oriented coordination, including rapid consensus, clear labor division, and efficient integration of feedback. Conversely, social cohesion is operationalized through interpersonal bonding and team commitment, evidenced by inclusive participation, supportive communication during disagreements, and resilience during technical or creative breakdowns (Table 1). In addition, individual attraction to the group is reflected in students' expressed satisfaction working with teammates and their preference to continue collaborating, while group integration is reflected in perceived unity and “we-ness” during task coordination. These construct-to-indicator links guided the selection of mixed-method data sources (screen recordings, cohesion notes, and questionnaires) and were used in the Results and Discussion to connect cohesion-related psychological processes to behavioral performance outcomes (writing quality and time-based efficiency).

2.2. AI and motivation

Recent studies across various disciplines demonstrate that AI chatbots, both custom-built and generic LLMs, significantly boost learner motivation, self-efficacy, and academic performance. In specialized fields like nursing (Chang et al., 2022) and public health (Lee et al., 2022), chatbots enhance achievement through personalized feedback and targeted training. Similarly, in EFL contexts, AI-powered tools yield substantial gains in writing proficiency and self-regulation (Apriani et al., 2024), while features such as “human-likeness” and social presence predict better oral outcomes (Wei et al., 2025). Broadly, AI chatbots are positive predictors of academic self-efficacy, a relationship further moderated by students' ability to self-regulate (Tanveer et al., 2024).

These individual-level gains extend to collaborative contexts through

Table 1
Contracts, indicators in SCW, and data sources.

Construct (GCT)	Definition in this study	Observable indicators in SCW	Data source(s)
Task cohesion	Shared commitment and coordinated effort toward the writing goal and successful task completion	Clear role enactment during drafting/revision; rapid consensus on ideas and wording; efficient revision cycles; prompt uptake of peer suggestions; fewer or shorter breakdowns in coordination (e.g., stalled discussion, repeated disagreement without resolution)	Screen recordings; cohesion notes; writing time/proofreading time metrics; writing scores
Social cohesion	Interpersonal bonding and willingness to maintain the group relationship during collaboration	Encouragement and positive socio-emotional talk; inclusive participation and turn-sharing; respectful conflict handling; supportive peer feedback; sustained engagement during difficulty (e.g., continuing to contribute rather than withdrawing)	Cohesion notes; screen recordings
Individual attraction to the group (ATG)	Individual members' positive orientation toward their team as a desirable group to belong to and work with	Statements of enjoyment or satisfaction with the team; preference for continued collaboration with the same teammates; perceived value/competence of teammates; expressed comfort communicating within the group	Cohesion notes; questionnaire items on perceived collaboration (if included); optional follow-up reflections (if collected)
Group integration (GI)	Perceived unity and “we-ness” in functioning as a team, including shared responsibility for outcomes	Coordinated decision-making routines; shared ownership language (we/our); stable collaboration routines across phases (planning → drafting → revising); collective responsibility statements (e.g., “our conclusion needs...”)	Cohesion notes; screen recordings

collective efficacy (Bandura, 1997), a group's shared belief in its ability to organize and execute actions to attain its goals. In interdependent tasks, tools that enhance perceived conjoint capability (e.g., AI as a reliable “team member”) can boost motivation by fostering sustained effort and reducing demotivation due to overload or conflict. Emerging work on AI in collaborative settings supports this: chatbots promote group cohesion via emotional support and dialogue facilitation (Kim & Lim, 2025), while hybrid human-AI patterns show cognitive/socio-emotional interactions that sustain engagement (Feng, 2025).

2.3. Key unresolved issues in group cohesion, AI assistance, and SCW literature

Despite advancements in the extant literature on research on technology-enhanced collaborative writing (CW) in L2 contexts, three critical gaps remain:

Methodological and contextual disparities: Studies over-rely on self-reporting and discourse analysis rather than unobtrusive “trace data” (Zamecnik et al., 2022). Furthermore, research favors online/asynchronous settings, leaving a gap in understanding how face-to-face synchronous collaborative writing (SCW) and different tools, like shared documents versus GenAI, impact real-time negotiation and cohesion (Tao et al., 2022; Zhang & Chen, 2022).

Neglected affective mechanisms: While CW improves text quality and performance, internal drivers like motivation and self-efficacy are rarely measured (Li & Zhang, 2023). Only isolated studies, such as Ebadijalal and Moradkhani (2025), explicitly report motivational gains.

Isolated AI research: While AI chatbots have been shown to boost individual self-efficacy and reduce anxiety across various fields (Apriani et al., 2024; Chang et al., 2022), these benefits haven't been studied in group SCW. It remains unclear if AI tools (like Kimi AI) will amplify collective efficacy or disrupt human interaction.

These issues highlight the need for studies articulating how tool type shapes cohesion-motivation-performance pathways in real-time SCW. To address this, we propose the explanatory model shown in Fig. 1, which integrates GCT with social interdependence theory and collective efficacy.

The research answers three key questions: (1) How do AI chatbots versus online resources influence performance in offline in-class SCW? (2) What are students' perceptions of these writing support tools? (3) How does resource type affect team cohesion during the SCW task?

3. Methodology

3.1. Participants

Two sophomore classes, comprising sixty-three students, including 33 (female = 25; male = 8) in the Experimental Cohort (EC) and 30 (female = 23; male = 7) in the Control Cohort (CC), participated in the study. At the beginning of the semester, the administration randomly assigned the students to two classes by alternating students' registration numbers. The EC and CC were randomly assigned by flipping a coin, and students were randomly assigned to teams in both classes. The EC and the CC each had 6 teams, comprising 5–6 members. Baseline scores were obtained pre-study, training was provided, and all ethical considerations were met. The same classroom was used with identical environmental conditions and infrastructure, and each student used their individual PC. In a short survey, no students reported using AI during SCW, and use of web-based resources was minimal.

3.2. Study design

The study took place during the six weeks of the new semester and focused on revising descriptive writing, the previous semester's course content. Students were familiar with the entire writing process (brainstorming, structuring, paragraph development, and draft proofreading), collaborative work, and the use of online resources. Students' extensive prior group collaboration experience (multiple tasks in the previous semester involving brainstorming, structuring, paragraph development, and drafting) minimized novelty or Hawthorne effects. Post-intervention differences in performance, motivation scale, cohesion notes, and perceptions indicated sustained impacts rather than transient novelty influences.

3.2.1. Study objectives

The lesson aimed to revise the descriptive writing skills learned in the previous semester and reinforce them by focusing on adjectives, word choice, collocations, and overall content and organization. Another objective was to train students to solicit assistance (from web-based resources and AI) with writing, particularly in vocabulary, collocations, tone, and translation.

3.2.2. Previous training and baseline

In the previous semester, students completed multiple collaborative descriptive writing tasks (brainstorming, structuring, paragraph development, drafting, and revision) both in and out of class. To maximize discussion time, teams typically assigned one member to take notes while others contributed ideas and language support (e.g., searching for terminology and inserting it into the draft after group discussion). Guidelines ensured equal participation and flexible role assignments. Students were also trained to use EV Recorder software for screen capture, enabling them to record, save, and submit MP4 files to the instructor. Prior training emphasized team organization and writing processes, with minimal focus on technological supports (web-based tools or AI), and students demonstrated strong familiarity with group collaboration. Therefore, to address gaps in the literature regarding the impact of technology on group cohesion in collaborative settings, this study incorporated two key additions: first, students practiced working synchronously on a cloud-based shared document, allowing team members to discuss and write simultaneously. Second, the EC was introduced to Kimi AI to assist them during SCW. The CC consulted online resources, which they had rarely used in the previous semester.

Baseline assessment occurred at the end of Week 2. After curriculum revision and reinforced team training, both EC and CC groups collaborated to produce a 500-word descriptive essay (assessed against the Appendix 1 criteria). Independent-samples *t*-tests revealed no significant or practical group-based pre-intervention differences in writing performance. Students also completed the pre-intervention Motivated Strategies for Learning Questionnaire (MSLQ) to establish baseline motivation and self-efficacy levels. The MSLQ (Appendix 3) was selected for its robust validation in higher education and L2 writing contexts (Huang, 2008), with subscales that directly map to intrinsic/extrinsic motivation and self-efficacy for learning/performance.

3.2.3. Tool-based training and pilot

AI training was provided in Weeks 3 and 4. Students in the EC were introduced to Kimi AI, a Chinese-designed LLM (Chen et al., 2024) with characteristics similar to ChatGPT. Kimi was selected for its advanced natural language processing capabilities, aligning with the study's objective to explore how AI can assist with vocabulary, collocations, and tone adjustments in descriptive writing. Kimi's design as a localized language model tailored for Chinese users increased the tool's suitability.

First, an expert was invited to teach students prompting. In particular, they learned to write prompts to obtain the correct collocations, words, phrases, synonyms, antonyms, and tone, and translate words and phrases into English. Each student consulted the chatbot on their individual PCs, then, while recording their screens, copied and pasted or wrote their findings on the shared cloud-based platform to improve the essay. Two trained assistants monitored the students during training to ensure consistency and uniformity. They completed a checklist and documented any problems that the instructor addressed. In-class training was followed by homework, with students instructed to report any challenges they encountered through the class WeChat instant messaging platform.

Meanwhile, students in the CC learned to use TWRs (e-dictionaries, specialized dictionaries, bilingual dictionaries, corpora, and machine translation systems). A list was created for students to consult when necessary. The hired assistants also monitored them during this period. The pilot study took place in Week 5. Students in both classes continued

to work collaboratively, sourcing information from Kimi AI (EC) and TWRs (CC). They brainstormed ideas, structured them, and wrote the descriptive essay paragraphs collectively on the shared cloud-based platform, as outlined in Fig. 2.

Teams sat around a table and collectively wrote the essay on Kdocs, a Tencent-developed cloud-based document-sharing platform, with the EC and CC students reinforcing their ability to use Kimi AI and TWRs, respectively. The trained assistants monitored the students and completed a checklist as the experiment progressed. Students in both classes recorded their screens. Post-experiment MLSQ and perception questionnaires were completed, and students were provided with data on team cohesion.

As shown in Fig. 3, the intervention occurred in Week 6. It was monitored and controlled by the hired assistants, who, again, completed a checklist as the experiment progressed. Students in both classes recorded their screens. Post-experiment MLSQ and perception questionnaires were completed, and students were provided with data on team cohesion.

3.3. Data collection and instruments

To answer the research questions, five data sources were gathered, including typing speed, total writing time, writing performance, team cohesion notes, and motivation and perception.

To account for potential differences in efficiency that could influence task completion times, typing speed was first measured via www.typingtest.com before students worked collaboratively on cloud-based shared documents to write and submit their essays as Word files. Total writing time was captured via screen recordings and divided into essay writing time (EWT) and essay proofreading time (EPT), while final team performance was evaluated based on assessment criteria (Appendix 1), including content, organization, descriptive language, mechanics, collocation, word choice, engagement, and sensory details.

3.3.1. Motivation and perceptions of support received

Post-study, students were allocated 25 min to complete two questionnaires and provide qualitative data on team cohesion. They first filled the post-intervention MSLQ (Pintrich et al., 1993), a well-validated instrument assessing motivation (intrinsic/extrinsic goal orientation, task value), expectancy (control beliefs, self-efficacy for learning/performance), and affect (test anxiety), with established reliability in higher education and L2 contexts (Jackson, 2018; Pintrich et al., 1993).

Additionally, they completed a short perception questionnaire (10 items, 5-point Likert scale plus open-ended questions) adapted from validated tools on collaborative learning and AI integration (Troussas et al., 2023). Three educational experts confirmed the content's validity, and it was piloted and refined using 40 students from other classes. Its reliability was high (Cronbach's $\alpha = 0.87$), and it captured quantitative/qualitative insights into tool perceptions. Data collection was closely monitored to maintain consistent conditions.

3.3.2. Team cohesion

Team cohesion data were gathered post-intervention. It focused on communication and collaboration, roles and responsibilities, conflict resolution, motivation and engagement, and feedback. Students provided information individually in a Microsoft Word template. They were monitored by the hired assistants to ensure that each student worked independently. The data was emailed to the instructor for further analysis.

3.4. Data analysis

After ensuring ethical data integrity, statistical analysis revealed that typing speed had no significant impact on the study's brainstorming or writing outcomes. Researchers employed *t*-tests and Pearson correlations to rigorously compare groups and explore linear relationships

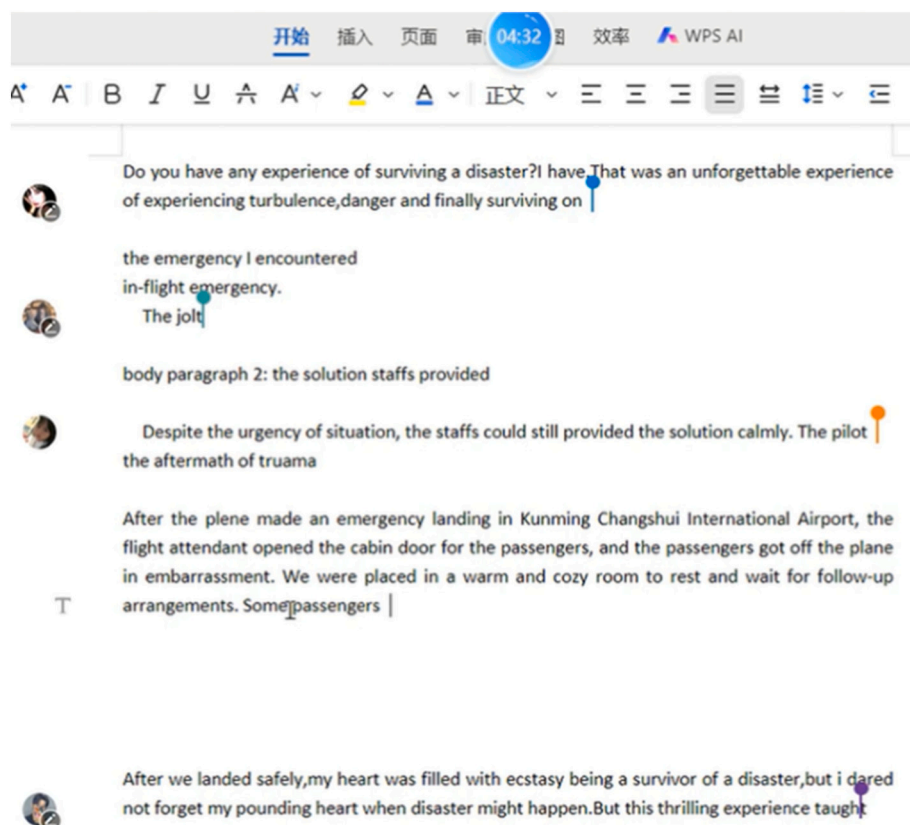


Fig. 2. Screenshot of SCW in the cloud-based platform.

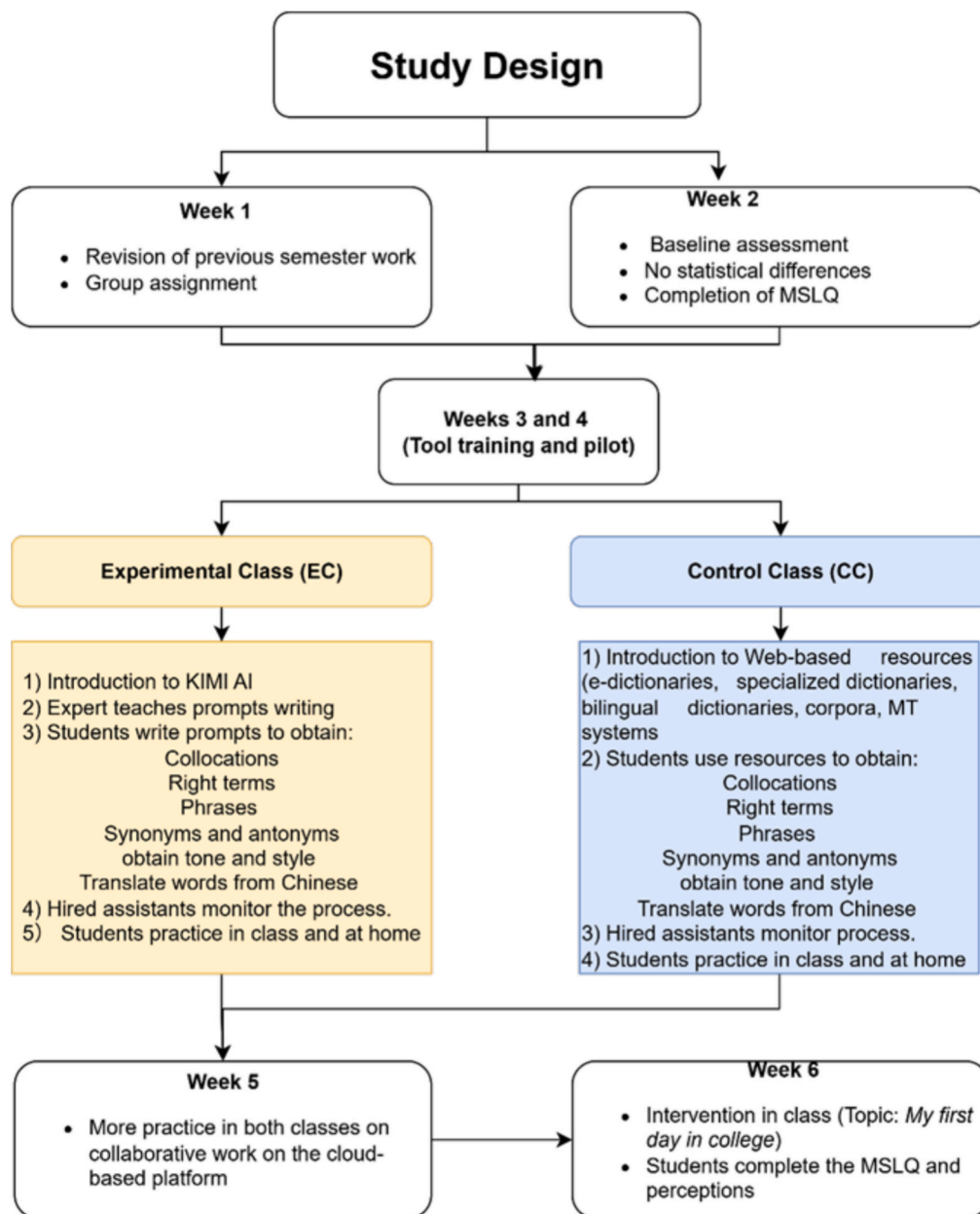


Fig. 3. Study design outline.

between variables.

RQ1: How do the AI chatbots and TWRs influence students' SCW performance?

To answer this question, we analyzed data on SCW time (i.e., essay writing time (EWT) and essay proofreading time (EPT)) and final assessment scores. EWT and EPT data were obtained from screen recordings using VNote (Tekwa, 2024), video analysis software, by two coders (Inter-coder reliability: Cohen's Kappa, $k = 0.89$). The time sequences for EC and CC were extracted, downloaded into MS Excel, and compared (Independent t -tests and descriptive statistics) on SPSS.

Performance was measured using essay assessment scores, which were blindly evaluated by three assessors. The assessment was based on the criteria outlined in Appendix 1, and scores were computed using SPSS. Three experts blindly assessed the tasks (inter-rater reliability was 0.93), indicating a high level of agreement among the three raters (Koo & Li, 2016). The performance of the two classes was compared on SPSS

(independent-samples t -test). Furthermore, the screen recordings were analyzed to identify patterns in the writing process that accounted for variation across the respective datasets.

RQ2: What were students' perceptions vis-à-vis their writing resources?

The questionnaire responses unveiled the perceptions (effectiveness, ease of use, teamwork, confidence, and satisfaction) of EC and CC students. Distributive data were obtained to understand differences among the various constructs and the datasets were compared on SPSS (Independent t -test). Furthermore, Pearson's correlation coefficients were computed to assess the relationships among perception, motivation, and performance.

RQ3: How did the writing resource influence students' motivations and group cohesion?

Motivation was assessed by analyzing MSLQ pre- and post-intervention data. Data was analyzed using ANCOVA test with all prior normality tests conducted to ensure robustness of outcomes. Pearson correlations were used to determine the relationships among the various variables investigated.

Qualitative data analysis: Students' open-ended notes and reflections on team cohesion and intra-team dynamics were analyzed using thematic analysis in NVivo (Version 12). Guided by prior work on group cohesion and collaborative writing, we first developed a deductive coding framework with five overarching categories: communication and collaboration, roles and responsibilities, conflict resolution, leadership and decision-making, and motivation and engagement. Two trained coders piloted this framework on approximately 15% of the data, allowing additional inductive codes to emerge where needed, and then refined the codebook (Appendix 2) through discussion. Then they independently coded all responses at the sentence- or clause-level. Inter-coder reliability was $\kappa = 0.89$, indicating substantial agreement; remaining discrepancies were resolved through consensus. NVivo's query functions were then used to compare the frequency and distribution of themes between the EC and CC groups and to select representative excerpts.

3.5. Ethical considerations

The study received approval from the university's research ethics department before initiation. Students provided informed consent, and privacy was prioritized by anonymizing all collected data, including screen recordings. Participants were informed about the study's purpose and their right to withdraw at any time without penalty.

4. Findings

4.1. RQ1: does AI improve speed and performance during SCW?

It should be cautioned that the observed effect sizes in this study are unusually large for educational research (e.g., Cohen's $d = 2.43$ for overall writing score; $\eta^2 = 0.888$ for post-task motivation). These magnitudes should be interpreted in light of the highly controlled, low-variability classroom context: a single institution, homogeneous English-major participants, identical tasks and environment, random class assignment, and intensive monitoring. With only six teams per condition, small-sample effect sizes are sensitive to reduced within-group variance. While the practical differences (17-minute time savings, 9-point score advantage) are meaningful, we caution against overgeneralization beyond similar tightly structured L2 SCW settings.

4.1.1. The EC had less EWT and EPT

Teams in the EC (Kimi AI) completed the task substantially faster than teams in the CC (TWRs). Due to the small number of teams ($n = 6$ per condition), both parametric and non-parametric tests were conducted.

Independent-samples t -tests were conducted to compare collaborative groups in the CC and EC on total writing time and total proofreading time. Teams in the CC required substantially more time to complete the essay-writing phase than those in the EC, with the difference highly

statistically significant ($p < .001$). Variances were homogeneous (Levene's test $p = .624$), and the effect size was exceptionally large (Cohen's $d = 12.61$; Hedges' $g = 11.64$). This efficiency gain in the AI condition apparently allowed teams to allocate more cognitive resources to content refinement rather than prolonged drafting.

In the proofreading phase (Table 2), the CC spent considerably more time than the EC ($p < .001$), again with homogeneous variances (Levene's test $p = .635$) and a very large effect size (Cohen's $d = 7.02$; Hedges' $g = 6.48$). The reduced proofreading burden in the AI group likely stemmed from more accurate and consistent initial drafts supported by the chatbot's uniform suggestions.

As a robustness check due to the small sample size, Mann-Whitney U tests were performed. Results fully confirmed the t -test findings, showing complete separation between groups with no rank overlap:

- Total writing time: $U = 0.00$, $Z = -2.882$, $p = .004$ (exact $p = .002$)
- Total proofreading time: $U = 0.00$, $Z = -2.887$, $p = .004$ (exact $p = .002$)
- Total time: $U = 0.00$, $Z = -2.903$, $p = .004$ (exact $p = .002$)

Mean ranks were 3.50 for the EC and 9.50 for the CC on all three variables. These results indicate that Kimi AI dramatically improved task efficiency. The practical difference was approximately 17 min faster per team overall. Findings should be interpreted in the context of the highly controlled classroom environment and small number of teams.

Video evidence: Statistical analysis showed no significant correlation between EWT and EPT in the CC ($p = .08$) and EC ($p = .34$). However, our analysis of individual group videos revealed two notable factors: The EC's superior efficiency stemmed primarily from Kimi AI's centralized interface, which provided uniform suggestions and eliminated the fragmented workflows characteristic of TWRs. Screen recordings showed that CC teams frequently switched between multiple tabs and sources, leading to longer pauses and redundant searches. In contrast, EC teams reached consensus more rapidly, allocating the time saved to content refinement rather than to information gathering. These behavioral patterns directly corroborate the quantitative time differences and set the stage for the cohesion and motivation findings presented below.

4.1.2. Overall, there were inter-class and intra-class performance differences between EC and CC

The descriptive statistics show a clear directional pattern favoring EC. Mean scores in CC ranged from 8.03 to 81.12, whereas EC means ranged from 8.50 to 90.28. Standard deviations were generally small in both groups (CC: 0.22 to 2.66; EC: 0.17 to 2.17), indicating relatively tight clustering within groups.

Inferentially, two-tailed p values ranged from <0.001 to 0.091 across categories. The strongest evidence (smallest p -values) was observed for Total Score, Use of Descriptive Language, and Collocation/Word Choice, with additional significant differences for Sensory Details and Content/Organization ($p = .01$ to < 0.001). By contrast, Language & Mechanics and Engagement & Voice showed weaker statistical separation in this dataset (p -value above 0.05).

For effect-size magnitude, Hedges' g values (small-sample corrected) ranged from about -5.00 to -1.68 for the clearest EC advantages

Table 2
Differences in the overall EWT and EPT between EC and CC.

Variable	Group	N (groups)	Mean	SD	t	df	Lower	Upper	p	d
EWT (writing time)	CC	6	1545.0	56.12	21.841	10	590.4241	724.5759	<0.001	12.61
	EC	6	887.5	47.83	21.841	9.755	590.1947	724.8053		
EPT (proofreading time)	CC	6	925.0	44.15	12.153	10	287.8744	417.1256	<0.001	7.02
	EC	6	572.5	55.66	12.153	9.509	287.4188	417.5813		

Note. These comparisons are at the team level ($n = 6$ teams per condition), resulting in a small effective sample size for effect size calculations.

(Descriptive Language, Collocation/Word Choice, Total Score, Sensory Details, Content/Organization), while effects were closer to -1.14 to -1.00 for the less stable domains (Engagement/Voice, Language/Mechanics). This mirrors the significance trend and supports a pattern of stronger short-term gains in descriptive/lexical performance and overall score than in all rubric components equally.

Overall, the statistical trend is consistent with $EC > CC$ across means, mostly low within-group dispersion, mixed but generally strong inferential support, and the largest corrected effects in descriptive and lexical writing domains, plus total performance.

For Language and Mechanics and Engagement and Voice, evidence was less conclusive in the reported models (Table 3), with intervals crossing zero. Given the six-week intervention and small number of collaborative groups, these results are best interpreted as short-term, domain-specific improvements, especially in descriptive quality, word choice, and overall writing performance, rather than broad claims of uniform improvement across all writing dimensions.

Intra-class performance

Performance varied by category, with no single team leading everything. In the CC, Team 4 showed the strongest overall profile, while Team 2 also stood out in Organization and Sensory detail. However, strong technical/descriptive scores did not consistently align with stronger voice. In the EC, the same uneven pattern appeared: Team 4 led consistently, Team 5 led descriptive language, and Team 3 ranked lowest across several core areas.

The correlation patterns were strong and positive. In CC, Collocation/Word Choice was strongly associated with Use of Descriptive Language ($r(6) = 0.86, p = .02$), and Content & Organization was strongly associated with Sensory Details ($r(6) = 0.82, p = .04$). In EC, Content & Organization was very strongly associated with Language & Mechanics ($r(6) = 0.92, p = .01$), with an equally strong association reported for Engagement & Voice ($r(6) = 0.92, p = .01$). Overall, related writing dimensions improved together.

Our screen recording analysis unveiled two fundamental practices that explained the differences between EC and CC performance: First, performance in Content & Structure was negatively impacted when team members deviated from their agreed-upon outlines during the actual writing process. For instance, CC Team 3 initially planned a paragraph detailing general cabin chaos during turbulence, yet the final draft shifted focus entirely to screaming children. Second, the indiscriminate use of adaptive technologies during proofreading frequently compromised semantic and grammatical quality. For instance, EC Team 2 mistakenly accepted a suggestion that changed “airport terminal” to “airport terminator.” Other grammatical errors across both groups included CC Team 4 using “decision” instead of “decided” and EC Team 5 using “accept” instead of “accepted,” because students reflexively clicked incorrect automated prompts.

Table 3
Intra-class differences for EC and CC.

Variable	Groups	N	Mean	SD	t	df	p	Lower	Upper	d	g
Content and Organization	CC	6	17.50	0.75	-3.151	10	0.01	-2.1312	-0.36547	1.82	1.68
	EC	6	18.75	0.61		9.604		-2.13616	-0.36051		
Use of Descriptive Language (20pts)	CC	6	15.31	0.85	-9.376	10	<0.001	-4.15846	-2.56154	5.41	5.00
	EC	6	20.20	0.94		5.776		-4.24517	-2.47483		
Language & Mechanics (25pts)	CC	6	20.61	0.76	-1.873	10	0.091	-2.12747	0.18414	1.08	1.00
	EC	6	21.61	0.99		9.222		-2.14082	0.19748		
Collocation and Word Choice (15pts)	CC	6	12.42	0.17	-9.929	10	<0.001	-2.62609	-1.59391	5.73	5.29
	EC	6	14.22	0.17		6.193		-2.62609	-1.59391		
Engagement & Voice (10 pts)	CC	6	8.03	0.22	-2.141	10	0.058	-0.96604	0.01937	1.24	1.14
	EC	6	8.50	0.45		6.964		-0.99677	0.05011		
Sensory Details (10pts)	CC	6	7.55	0.50	-3.863	10	0.003	-1.57942	-0.42391	2.23	2.06
	EC	6	8.55	0.37		9.411		-1.58436	-0.41898		
Total scores	CC	6	81.12	2.66	-6.541	10	<0.001	-12.2871	-6.04294	3.78	3.49
	EC	6	90.28	2.17		9.621		-12.3038	-6.02617		

4.2. What were students' perceptions of AI and TWRs during SCW?

We investigated differences between Kimi AI and TWRs in several categories: effectiveness, ease of use, team communication, and collaborative work enhancement. In addition, we gauged users' confidence levels and the challenges encountered. Overall, the EC had a higher perception of the AI chatbot than the CC did for TWRs (Table 4).

Data analysis showed that the EC had higher perceptions of their AI tool regarding its effectiveness ($M = 4.73$; $SD = 0.45$), ease of use ($M = 4.79$; $SD = 0.42$), team communication ($M = 4.70$; $SD = 0.47$), and confidence ($M = 4.76$; $SD = 0.44$), i.e., four of the six items assessed. In contrast, no statistically significant differences were found between Kimi AI and TWRs in satisfaction and group work enhancement. These findings indicate that, though the EC had a higher overall perception of AI in SCW, the choice of tool or resource did not affect members' satisfaction or the enhancement of collaborative work.

4.3. Motivation, team cohesion, and correlation with performance

We sought to understand the extent to which intra-team dynamics influenced students' performance in the CC and EC. Therefore, we assessed two sets of data: the motivation questionnaire data and the qualitative data presented in each student's notes. Establishing a correlation between motivation and performance was considered equally relevant to understanding intra-team dynamics.

4.3.1. Motivation (ANCOVA analysis)

A preliminary analysis indicated that the EC and CC did not differ on pre-test motivation, $F(1, 61) = 0.59, p = .446, \eta_p^2 = 0.01$. An ANCOVA was then conducted to compare post-test motivation between the EC and the CC, controlling for pre-test motivation. The assumption of homogeneity of regression slopes was met because the $\text{Group} \times \text{Pre-test motivation}$ interaction was not significant, $F(1, 59) = 0.04, p = .843$, partial $\eta_p^2 = 0.001$. However, Levene's test was significant, $F(1, 61) = 5.76, p = .019$, indicating unequal error variances. In the final ANCOVA model (without the interaction term), pre-test motivation was not a significant covariate, $F(1, 60) = 0.07, p = .791$, partial $\eta_p^2 = 0.001$, but the adjusted group effect on post-test motivation was significant and very large, $F(1, 60) = 475.18, p < .001$, partial $\eta_p^2 = 0.888$. Descriptively, post-test motivation was higher in the EC (unadjusted $M = 6.48, SD = 0.37$) than in the CC (unadjusted $M = 4.94, SD = 0.10$), suggesting that students who used Kimi AI during writing reported substantially greater motivation at post-test than students using other online resources, as outlined in Table 5.

To better understand the differences in motivation, especially for the CC, we examined students' qualitative data (notes) on team dynamics, focusing on communication, roles and responsibilities, conflict resolution, leadership and decision-making, and motivation and engagement.

Table 4
Perceptions of EC and CC on AI chatbot and TWRs.

	Class	N	Mean	SD	df	P-value	Effect size	Comment
Effectiveness of AI or TWRs	EC	33	4.73	0.45	61.00	0.001	2.82	Significant
	CC	30	3.27	0.58	54.59			
Ease of use of AI or TWRs	EC	33	4.79	0.42	61.00	0.001	1.16	Significant
	CC	30	3.97	0.93	39.34			
AI or TWR enhancement of collaboration among teammates	EC	33	4.64	0.55	61.00	0.234	0.30	Non-significant
	CC	30	4.47	0.57	59.87			
Intra-team sharing of ideas	EC	33	4.70	0.47	61.00	0.001	1.22	Significant
	CC	30	3.97	0.72	48.99			
Confidence using AI or TWRs	EC	33	4.76	0.44	61.00	0.001	1.31	Significant
	CC	30	3.83	0.91	40.65			
Satisfaction with AI or TWRs	EC	33	4.27	0.98	61.00	0.575	-0.14	Non-significant
	CC	30	4.40	0.81	60.56			

Table 5
ANCOVA results of pre- and post-motivation EC and CC.

Descriptive statistics and ANCOVA results for motivation (EC vs CC), controlling for pre-test motivation				
Panel A. Descriptive statistics (post-test motivation)				
Group	n	Post-test M	Post-test SD	
Experimental class (EC: Kimi AI)	33	6.48	0.37	
Control class (CC: other online resources)	30	4.94	0.1	
Panel B. Assumption tests and ANCOVA effects				
Test/effect	df	F	p	Partial η^2
Levene's test (error variance equality)	1, 61	5.76	0.019	—
Group $\times$ Pre-test motivation (slopes test)	1, 59	0.04	0.843	0.001
Final ANCOVA group effect (EC vs CC) ^a	1, 60	475.18	<0.001	0.888
Pre-test motivation covariate ^a	1, 60	0.07	0.791	0.001

Note. EC = experimental class (used Kimi AI during writing). CC = control class (used other online resources). Levene's test was significant, indicating unequal error variances. The Group $\times$ Pre-test interaction was not significant, supporting the homogeneity of regression slopes assumption.

^a These rows come from the ANCOVA model without the interaction term, which is typically reported when the interaction is non-significant.

4.3.2. Correlations

We examined Pearson's correlation coefficients among the various variables, i.e., motivation scores, EWT and EPT, perception of resources, and performance, to determine relationships that could provide further insights into the students' psychology and writing task. Regarding the EC, we found a negative correlation between performance on the writing task and proofreading time ($r(6) = -0.93, p = .003$), suggesting that using Kimi AI led to more efficient or effective writing, reducing the need for extensive proofreading. The results might also mean students were more confident in their writing and relied less on lengthy proofreading, possibly due to the accuracy of information provided by AI. In contrast, for the CC, we found a strong positive correlation between performance on the writing task and perceptions of their resources ($r(6) = 0.83, p = .04$), suggesting that confidence in or familiarity with TWRs enhanced their ability to use them, leading to better task outcomes. No other correlations were found for the two classes.

4.3.3. Team cohesion notes

Notes analysis showed similarities and differences in team cohesion that supported the quantitative findings. Kimi AI streamlined discussions for the EC and reduced resource search time, enhancing task focus. In contrast, the CC faced information overload from multiple sources, leading to prolonged discussions and disagreements that impacted

overall performance. Data revealed the following themes:

Communication and collaboration: Both EC and CC students had easy access to information and could communicate the information among group members without difficulty. However, the EC had easier access, given that they used an AI chatbot to obtain information, which was faster and uniform. For instance, Participant 23 (P23) in the EC explained that the AI delivered consistent information to all group members, eliminating the need for individual repeated searches. That increased and eased the process to obtain the required details and reach agreement on them. The ability to agree on AI-provided information presumably increased team spirit, reinforcing engagement and motivation. The CC data indicated that team members had easy access to information; however, the many information sources led to tensions and disagreements. For example, as P2 in the CC mentioned, “*We had much information from many dictionaries, so we sometimes discussed for a long time to find the right words or agree...Some people always searched Youdao dictionary and some always searched others. So, we had to decide which information to use. But, in the end, we always agreed.*” Even though teams finally agreed, the debates and disagreements appeared to have prolonged their discussions, as evidenced by the long pauses in some CC teams' screen recordings. These accounts illustrate a key mechanistic difference: Kimi AI's uniform and rapid responses facilitated swift consensus on content, thereby reducing interpersonal friction and strengthening task cohesion through alignment on shared goals. In contrast, the multiplicity of web sources in the CC generated divergent information, prolonging debates and potentially undermining task cohesion.

Roles and responsibilities: Our data analysis showed that, to a large extent, teams in both classes assigned specific roles to different team members. For instance, each of the 12 groups (6 in each class) chose one person to write the paragraph, while other members inserted or modified sentences when necessary. That means there was one writer and several researchers whose contributions were limited to looking for specific information and updating the paragraph. However, while the EC process was more streamlined and seamless, the CC process was less so. Tensions arose after members indiscriminately updated the paragraph. P15 remarked that “*Our group members found so much information sometimes, and they updated the passage. Then some other members who found alternative information also updated the paragraph. It was confusing because everyone was writing at the same time.*” To resolve the issue, the team decided to discuss and agree on all changes prior to any modifications. This way, members only modified the paragraph after the group deliberation and received the green light.

Conflict Resolution: According to EC, there were a few conflicts needing resolution. The data analysis showed that, compared with EC, CC had more conflicts to resolve. The conflicts were mostly due to disagreements about which resources to consult and personal

preferences that inhibited group work. For instance, P18 preferred Bing for information searches. During group work, they were assigned to use the lesser-known [Thefreedictionary.com](https://www.thefreedictionary.com). Then the resources found were not considered useful. So, P18 spend valuable time convincing the group. This example showed that teams in the CC had more challenges resolving conflicts and focusing on the quality of the paragraph. In contrast, the EC teams had fewer conflicts, given that they used one tool that provided them with the same information.

Additionally, AI provided very precise information to improve the quality of the writing. Though members often disagreed over the choice of specific words and phrases (*"Our arguments were mostly about which word or sentence to use"* (P26)), the EC appeared to have fewer arguments. P28 confirmed this, stating: *"Each time we asked a question, we got an accurate answer. For instance, when we asked for help with vocabulary, sentence structure, or tense, Kimi AI understood what we needed and gave us the true answer. For me, it was always a better answer than I could think of. So we agreed easily. Kimi AI was like our team member too; that is how we felt."* The lower incidence of conflict in the EC highlights how AI's consistent, accurate outputs minimized disagreement over resource quality, directly supporting GCT's emphasis on attraction to group tasks via reduced barriers.

Motivation and engagement: Students in the EC benefited from a streamlined process and support from the chatbot, leading to a boost in motivation. Their notes indicated that motivation was directly linked to AI use, specifically to the quality and speed of responses, their respective roles, and the perception that Kimi AI was an additional team member. As P14 remarked, *"My motivation was mostly due to the AI. It was like our team member because we could talk to him and ask him questions. Also, my teammates were nice, and we had a good discussion."* Additionally, motivation was linked to members' familiarity with each other, especially outside the classroom. P1 explained that their motivation stemmed from AI's performance, their team's productivity, and their closeness with most team members. *"I live in the same dorm with XXX, and we spend a lot of time together. We agree on most things and we encourage each other with studies. So, we always motivate each other."* The data suggest that the motivation translated into engagement, as the EC spent less time on the task and outperformed the CC.

The CC data indicated that teams remained motivated during the activity due to familiarity among members and the volume of information they could mine online. As P25 in the CC stated, *"I was able to use it to find a lot of synonyms and useful phrases that improved our writing. And my team used the information. That made me happy and motivated."* However, this feeling was not shared by all the students who searched for information. When the information was not used, members tended to be demotivated. For instance, P16 stated, *"My team members were not very cooperative. I looked for a lot of information, such as tenses and words from the dictionary and Linguee. I also noticed some Chinglish and fixed it. But, they all rejected my answers. And then we got a low score. I was very frustrated."* The data suggest that while the assignment of roles was a positive action, team members' motivation and engagement depended on whether their effort paid off. In other words, the challenges of searching for and evaluating multiple resources can be demotivating. Furthermore, some members of the CC felt overwhelmed or confused by the abundance of information they sought. P15 wrote, *"I was writing the paragraph, and others were searching for information. They found too many alternatives for everything, and I was not sure which was the best option. Then, everyone wanted their information added to the paragraphs. So, I didn't want to continue writing."* EC narratives reveal AI as a perceived team member that boosted engagement and group cohesion, especially when students' contributions were validated by group members. Conversely, rejected or overwhelming inputs eroded motivation, illustrating how

resource inconsistency can disrupt collective efficacy.

Feedback: The EC's overarching response was that Kimi AI provided instant, accurate feedback on queries, acting as an additional team member rather than a support tool. The instant feedback meant that the teams in the EC were focused on organizing the writing task and improving its quality. As P13 maintained, *"The feedback from Kimi AI was fast and good quality. I think the AI chatbot gave us the answers we needed. So, we tried to discuss the paragraph structure, make the sentences better, and use Kimi as much as possible."* In addition to Kimi AI's feedback, EC teams believed peer feedback also contributed to improving the quality of their writing.

Similarly, the CC teams found that the online resources provided them with much useful information to improve writing quality. Students had more choices, more information, and many more options mined from online sources. However, some members complained about the slow and inconsistent feedback: *"I think the feedback from my teams was slow. They had to look up the words and expressions first; then, we used them to improve our writing. It was my task to write the paragraphs and I waited for contributions from my teammates. Sometimes, it was slow to come, so our improvement in writing was a little slow"* (P12).

These qualitative accounts of streamlined communication, reduced conflict, and enhanced task focus converge with the large efficiency and performance gains reported in 4.1, illustrating the mechanisms underlying the explanatory model.

4.3.4. Triangulation of cohesion findings

To ensure the validity of the results, the quantitative metrics were triangulated with behavioral traces from the screen recordings and qualitative data from individual students' notes. First, the quantitative speed advantages observed in the EC (EWT: $M = 24:20$; EPT: $M = 9:33$) directly align with the behavioral patterns captured in the screen recordings. While CC video traces showed fragmented workflows and prolonged pauses due to constant tab switching across disconnected TWR windows, the EC recordings showed rapid consensus building facilitated by Kimi AI's centralized interface. This behavioral trace mirrors the qualitative entries in the student notes, where EC participants highlighted "consistent information" and "faster agreement".

Second, the screen recordings and qualitative notes reveal divergent underlying causes for the performance variances across the groups. In the CC, errors and delays were driven by teams structurally deviating from their initial outlines during drafting (e.g., Team 3). In contrast, the EC's performance dips in specific rubric categories were driven by automation bias; screen recordings captured instances where teams reflexively accepted flawed automated prompts without peer negotiation.

5. Discussion

5.1. Summary of findings and contribution to GCT

The results confirm that Kimi AI improved task efficiency, writing performance, and post-task motivation, consistent with prior AI-supported writing studies (Chang et al., 2022; Li et al., 2025; Zhang & Chen, 2022). As reported in Section 4.3.3, however, the qualitative data revealed that GenAI moves synchronous collaborative writing from a predominantly negotiation-driven model, where groups debate divergent information from multiple sources, to a coordination-driven model, in which uniform, rapid AI support enables swift consensus and greater focus on joint text refinement.

This transition from negotiation to coordination explains the sharp divergence between the quantitative surge in post-task motivation and the social constraints documented in student reflections. While Kimi AI boosted collective efficacy by acting as an efficient, neutral team member who smoothed operational friction, it simultaneously truncated

the interpersonal dialogue vital to deep collaborative learning. From a GCT standpoint, these findings suggest that task cohesion and social cohesion are structurally decoupled in AI-mediated synchronous writing environments. Centralized AI platforms accelerate coordination-driven collaboration. They maximize task cohesion by aligning the group around rapid and automated consensus, but they simultaneously minimize the socio-cognitive conflicts that foster interpersonal bonds. Consequently, while GenAI optimizes the production process of a joint text, it risks thinning out the social fabric of the student group.

This study changes current understandings of AI in collaborative learning in two important ways. First, it demonstrates that AI does not merely enhance collaboration; it may qualitatively transform its underlying mechanism from negotiation-oriented to coordination-oriented. Second, it extends GCT by showing that task and social cohesion can be differentially affected (and partially decoupled) by the type of technological support. While AI boosts coordination, efficiency, and collective efficacy, its very efficiency may limit the relational depth traditionally associated with collaborative learning, a trade-off not fully captured by quantitative measures alone.

This coordination-driven model, while powerful for task performance, reveals a clear imbalance in its effects on the two dimensions of GCT. While AI strongly enhanced task cohesion through faster consensus, clearer role integration, and reduced coordination costs, its impact on social cohesion was more limited and in some cases negative. By providing rapid, uniform answers, Kimi AI reduced the need for extended peer debate, perspective-taking, and negotiation of alternatives. These processes are crucial for building interpersonal bonds, mutual trust, and a sense of “we-ness” (Carron et al., 1985). Several EC students noted that although the AI felt like “an additional team member,” the accelerated pace left less room for personal contributions, emotional support, and collective problem-solving. Consequently, while task cohesion and short-term performance improved dramatically, the depth of peer bonding and the quality of collaboration may have been constrained. This trade-off highlights that AI does not uniformly strengthen teamwork; it excels at coordination but may inadvertently weaken some relational aspects of collaboration.

5.2. Implications

The findings suggest that AI can be integrated into synchronous collaborative writing to improve efficiency, enhance selected aspects of writing quality, and sustain motivation. However, its use must be deliberately scaffolded to mitigate risks of over-reliance and reduced peer negotiation. Teachers should position GenAI tools as a support resource for planning, vocabulary and collocation work, tone adjustment, and initial drafting. To harness AI's coordination advantages while preserving the socio-emotional and critical dimensions of collaboration, the following concrete instructional strategies are recommended:

Mandatory justification of AI outputs: Require teams to keep a simple “AI Decision Log” (either in a separate column of the shared document or a short shared note). For every significant AI suggestion incorporated into the essay, students must briefly record (1) the original AI output, (2) their group's decision to accept, modify, or reject it, and (3) a one-sentence justification based on linguistic appropriateness, contextual fit, or stylistic goals. This practice encourages critical evaluation rather than passive acceptance and directly counters the reduced negotiation observed in the AI condition.

Structured disagreement tasks: After producing an initial AI-assisted draft, allocate 10–15 min for a deliberate “AI-free revision round.” During this phase, teams must generate and debate at least two alternative formulations (e.g., different descriptive phrases or sentence structures) without consulting AI, then collectively choose the strongest version. This strategy restores the negotiation-driven

processes that AI tends to shortcut, thereby supporting social cohesion while still benefiting from earlier efficiency gains.

Role-based collaboration protocols: Implement rotating roles within each team, such as “AI Prober” (responsible for formulating prompts and retrieving suggestions), “Critique Lead” (evaluating AI output for accuracy and relevance), “Integrator” (blending AI suggestions with peer ideas), and “Peer Validator” (ensuring all members' voices are represented). Rotating these roles every 15–20 min ensures active participation and prevents any single member (or the AI) from dominating the process.

Discussion checkpoints: Insert two mandatory peer-only pauses during the task, after outlining and after completing the first full draft, during which teams discuss content and language choices for 5–7 min without AI access. Only after peer discussion may they return to the chatbot for targeted support. These checkpoints help maintain interpersonal interaction and collective meaning-making.

When implemented together, these strategies enable instructors to leverage AI's ability to strengthen task cohesion and collective efficacy while safeguarding the relational processes that nurture social cohesion. The motivational benefits observed in this study are most likely to endure when students remain active authors who critically engage with AI output rather than passive recipients. Ultimately, the pedagogical value of GenAI in collaborative writing lies not merely in faster production but in its thoughtful integration into well-designed group processes that balance coordination efficiency with meaningful peer collaboration.

6. Conclusion

This study compared the use of Kimi AI and traditional web resources in offline, in-class synchronous collaborative writing and examined their effects on writing performance, learner perceptions, and team cohesion through both quantitative and qualitative data. Overall, the findings suggest that Kimi AI offered clear advantages in task efficiency, selected aspects of writing performance, and student motivation. The EC completed writing and proofreading more quickly, produced stronger texts across assessed categories, and reported greater ease of use, confidence, and engagement. By contrast, students using traditional web resources experienced more delays, longer pauses, and greater difficulty coordinating information during collaboration.

At the same time, the findings point to a more nuanced role for AI in collaborative writing. Although AI appeared to support smoother task execution and stronger task cohesion by helping groups access shared information quickly and reach consensus more efficiently, its benefits were not uniform across all dimensions of collaboration. Students' accounts suggest that AI's speed and consistency may also reduce opportunities for extended discussion, comparison of alternatives, and peer negotiation. In this sense, AI may improve the flow of group work without necessarily deepening the quality of interaction or strengthening social cohesion to the same degree.

The study therefore contributes to CALL, SCW, and AI-in-education research in several ways. It offers an empirical comparison of a GenAI chatbot and traditional web-based resources in face-to-face L2 collaborative writing, an area that remains relatively underexplored. It also extends GCT by showing that AI may strengthen coordination and task-focused alignment more readily than interpersonal or socially grounded forms of cohesion. In addition, by examining motivation alongside performance and collaboration, the study addresses an affective dimension that has often received limited attention in collaborative writing research.

The findings do not support an unqualified endorsement of AI integration. Rather, they suggest that GenAI can be a valuable support for collaborative writing when used to enhance efficiency and provide timely assistance without displacing students' own judgment, interaction, and shared authorship. The pedagogical value of AI lies not simply

in making collaboration faster, but in integrating it carefully so that gains in performance and coordination do not come at the expense of cognitive engagement and meaningful peer exchange.

7. Limitations

This study has several limitations. First, while the student team assignment was random, the small sample size limits statistical power and generalizability, particularly for team-level analysis. Additionally, the Hawthorne effect may have occurred due to the controlled environment and the presence of monitoring assistants, which may have altered participant behavior. Second, the scope is limited by the use of a single AI platform (Kimi AI) and a focus on descriptive writing over one semester, which restricts the applicability of findings to other tools, writing genres, or long-term contexts. Third, the small sample size (n = 63; 12 teams total) limits statistical power and generalizability, particularly for team-level analysis. While the observed effect size was large (e.g., overall Cohen's $d = 2.43$), this figure should be interpreted cautiously; the high value likely reflects the restricted within-group variability of a homogeneous, highly controlled classroom setting rather than universal outcomes.

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Appendix 1. Assessment criteria for descriptive essay

Criteria	Excellent (4)	Good (3)	Satisfactory (2)	Needs improvement (1)	Score
1. Content and Organization	Clear, logical structure with well-defined introduction, body, and conclusion; ideas are cohesively linked.	Generally organized, with minor lapses in flow or structure; main ideas are mostly clear.	Some organization present, but lacks clarity or logical flow; main ideas may be undeveloped.	Disorganized; difficult to follow the essay's structure; main ideas are unclear.	20
2. Use of Descriptive Language	Rich, vivid language that effectively evokes imagery and engages the reader throughout the essay.	Good use of descriptive language, with some effective imagery, though it may lack depth in parts.	Limited descriptive language; some imagery present but not fully developed; relies on clichés.	Minimal use of descriptive language; fails to create imagery; overly simplistic.	20
3. Language and Mechanics	Virtually error-free; demonstrates a strong command of grammar, punctuation, and spelling.	Few minor errors that do not distract from overall readability.	Noticeable errors that occasionally hinder understanding; grammar and punctuation issues present.	Frequent errors that significantly disrupt readability; poor grammar and punctuation.	25
4. Collocation and Word Choice	Excellent use of collocations and precise word choices that enhance meaning and clarity.	Generally good collocation and word choice, with minor awkward phrases.	Some inappropriate or awkward collocations; word choice may lack precision.	Frequent misuse of collocations; vague or incorrect word choices that confuse the reader.	15
5. Engagement and Voice	Strong, engaging voice that draws the reader in and maintains interest throughout.	Engaging voice, though it may fluctuate or lack consistency at times.	Some engagement, but voice is weak or inconsistent; lacks personality.	Lacks a distinctive voice; fails to engage the reader; monotonous tone.	10
6. Sensory Details	Excellent incorporation of sensory details (sight, sound, smell, touch, taste) that enhance the description.	Good sensory details included, though some senses may be underutilized.	Few sensory details; relies heavily on one or two senses; lacks depth.	Lacks sensory details; fails to engage the reader's senses.	10

Appendix 2. Codebook for qualitative analysis on NVivo

Code	Definition	Example excerpt (paraphrased)
Communication_and_Collaboration	References to how team members exchanged information, shared ideas, coordinated writing, and worked together toward the task, including perceived clarity, efficiency, and openness of communication.	"We discussed each paragraph in the group chat before anyone wrote, so it felt like we were building the essay together."
Roles_and_Responsibilities	Descriptions of how tasks were divided or shared, how roles were assigned or negotiated (e.g., leader, writer, editor), and whether responsibilities were perceived as fair and clear.	"At the beginning, we decided one person would draft and the others would revise, which made our responsibilities clear."
Conflict_Resolution	Mentions of disagreement, tension, or conflicting ideas within the team, and how these issues were addressed or left unresolved. Includes both constructive and unproductive conflict.	"We had different opinions about the thesis, but we calmly explained our reasons and finally agreed on a better version."

(continued on next page)

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CRedit authorship contribution statement

Rong Han: Writing – review & editing, Writing – original draft, Visualization, Supervision, Project administration, Methodology, Investigation, Funding acquisition, Formal analysis, Data curation, Conceptualization.

Ethical considerations

This study was conducted in conformity with ethical guidelines stipulated in the Helsinki Agreement and with approval from the research department of Xi'an Medical University (No. XYLS2025146 of Jan. 12, 2025).

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

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NA.

(continued)

Code	Definition	Example excerpt (paraphrased)
Leadership_and_Decision_Making	Comments about who took initiative, guided the group, or coordinated decisions, and how decisions were ultimately made (e.g., majority vote, consensus, one dominant member).	“One teammate naturally became the leader and helped us decide the order of paragraphs when we could not agree.”
Motivation_and_Engagement	Statements about members' willingness to participate, effort, interest in the task, or emotional reactions that affected engagement (e.g., enthusiasm, anxiety, frustration).	“Using the AI suggestions made me more willing to try new ideas, so I felt more motivated to contribute.”
Overall_Team_Cohesion	Holistic judgments about how cohesive or “together” the team felt, including unity, belonging, and satisfaction with teamwork (often using terms like “cohesive,” “united,” “worked well together”).	“Overall, our team was cohesive; we respected each other's ideas and finished the task smoothly.”

Appendix 3. MSLQ questionnaire

Question	Description
Q1	In a class like this, I prefer course material that really challenges me so I can learn new things.
Q2	If I study in appropriate ways, then I will be able to learn the material in this course.
Q3	When I take a test I think about how poorly I am doing compared with other students.
Q4	I think I will be able to use what I learn in this course in other courses
Q5	I believe I will receive excellent grades in this class.
Q6	I'm certain I can understand the most difficult material presented in the readings for this course.
Q7	Getting a good grade in this class is the most satisfying thing for me right now.
Q8	When I take a test I think about items on other parts of the test I can't answer.
Q9	It is my own fault if I don't learn the material in this course.
Q10	It is important for me to learn the course material in this class.
Q11	The most important thing for me right now is improving my overall grade point average, my main concern in this class is getting a good grade.
Q12	I'm confident I can learn the basic concepts taught in this course.
Q13	If I can, I want to get better grades in this class than most of the other students.
Q14	I'm confident I can understand the most complex material presented by the instructor in this course.
Q15	When I take tests I think of the consequences of failing.
Q16	In a class like this, I prefer course material that arouses my curiosity, even if it is difficult to learn.
Q17	I am very interested in the content area of this course.
Q18	If I try hard enough, then I will understand the course material.
Q19	I have an uneasy, upset feeling when I take an exam.
Q20	I'm confident I can do an excellent job on the assignments and tests in this course.
Q21	I expect to do well in this class.
Q22	The most satisfying thing for me in this course is trying to understand the content as thoroughly as possible.
Q23	I think the course material in this class is useful for me to learn.
Q24	When I have the opportunity in this class, I choose course assignments that I can learn from even if they don't guarantee a good grade.
Q25	If I don't understand the course material, it is because I didn't try hard enough.
Q26	Understanding the subject matter of this course is very important to me.
Q27	I like the subject matter of this course.
Q28	I feel my heart beating fast when I take an exam.
Q29	I'm certain I can master the skills being taught in this class.
Q30	Considering the difficulty of this course, the teacher, and my skills, I think I will do well in this class.
Q31	I want to do well in this class because it is important to show my ability to my family, friends, employer, or others.

Data availability

Data will be made available on request.

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